

Supplementary Materials for: Predicting the long-term impact of AI assistants on collective intelligence

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Materials and Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (\text{S1})$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a prediction task (Wang et al., 2025). In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (\text{S2})$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player’s prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player’s strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (\text{S3})$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (\text{S4})$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (\text{S5})$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous work (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (\text{S6})$$

Second, when the group is assisted by the Chatbot AI, both the player’s initial prediction and the AI’s prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor’s contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (\text{S7})$$

where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A ’s revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (\text{S8})$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/N$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (\text{S9})$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging (Wang et al., 2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

99 To examine the effects of directly encouraging or discouraging AI use, we mod-
 100 ify the feedback and niche-expert structures by adding a term proportional to the
 101 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (\text{S10})$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (\text{S11})$$

102 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 103 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 104 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 105 the original feedback and niche-expert structures.

106 The balanced structure is defined as the weighted mean of the feedback and
 107 niche-expert payoff components, with balancing weight w :

$$\pi_A = (1 - w)Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - w\rho_{e_A}(Y_A^{\text{re}} - Y)^2, \quad (\text{S12})$$

108 where $w \in [0, 1]$ determines the relative contribution of the feedback and niche-expert
 109 components. We set $w = 0.5$ for all analyses unless otherwise specified, such that
 110 the two components contribute equally to the balanced structure. The sensitivity of
 111 the results to w is examined in Supplementary Fig. S7.

112 To examine whether incentives based on unrevised predictions (i.e., predictions
 113 made without incorporating AI predictions) could overcome the limitations of the
 114 other incentive structures, we defined an unrevised balanced structure. The unre-
 115 vised balanced structure is defined as the arithmetic mean of the feedback structure
 116 based on the unrevised collective prediction and the niche-expert structure based on
 117 unrevised individual predictions:

$$\pi_A = \frac{Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{2}, \quad (\text{S13})$$

118 where $\hat{Y}$ is the collective prediction based on unrevised individual predictions, and
 119 $Y_A(Y - \hat{Y})$ and $-\rho_{e_A}(Y_A - Y)^2$ represent the unrevised feedback and unrevised niche-
 120 expert components, respectively.

121 We further defined a combined balanced structure as the arithmetic mean of
 122 the balanced and unrevised balanced structures, with the balancing weight set to
 123 $w = 0.5$:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2 + Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{4}. \quad (\text{S14})$$

124 Simulations

125 All simulations are conducted using an agent-based model with 50 causal factors
 126 ($M = 50$) and 10,000 players ($N = 10,000$). The parameters defining the prediction
 127 target are sampled at the beginning of the simulations and subsequently held fixed.

Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization. Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

Players’ initial strategies are also sampled from predefined probability distributions. The distribution of individual belief c_A depends on the aggregation rule. Under simple averaging, each factor’s contribution is scaled by the proportion of players interested in that factor. We therefore sample beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player is sampled uniformly from the integers between 0 and 50, and reliance on AI is sampled from a uniform distribution on $[0, 1]$. To account for stochastic variation, we perform 30 independent simulation replicates with randomly initialized player populations for each condition presented in the main figures.

We assume that prediction events occur on a substantially faster timescale than social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs are derived for all incentive structures and are presented in the Supplementary Text.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (\text{S15})$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does not imitate player B but instead redraw all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4.

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

169 Collective accuracy is normalized relative to the baseline prediction α_0 . A nega-
170 tive value indicates that the collective prediction performs worse than the baseline.
171 Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the
172 evolved interests and beliefs of the AI-assisted population. In calculating interest
173 diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table S1).

Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

Model

Because Omniscient AI observes all M causal factors, its prediction for player A is given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S16})$$

where b_i represent factor-specific biases and ϵ represents stochastic error, following a Gaussian distribution with mean zero and standard deviation τ . A new stochastic error is generated independently for each player and prediction event.

Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S17})$$

In contrast to the AI prediction, player A 's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S18})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S19})$$

represents player A 's interest, c_A represents their belief about the effect of the corresponding factor, and β_A represents their reliance on AI. As in the main text, we set $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

198 Aggregation

199 Because the AI component of each revised prediction contains information about all
200 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (\text{S20})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (\text{S21})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon_j) + (1 - \beta_j) c_j X_{e_j}\} \quad (\text{S22})$$

201 To separate the contributions of AI and the players' initial predictions, we define the
202 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
203 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (\text{S23})$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon_j \quad (\text{S24})$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S25})$$

204 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
205 remaining after averaging, and γ_i represents the contribution of players' initial pre-
206 dictions to factor i . Using these definitions, the revised collective prediction can be
207 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S26})$$

208 Thus, the collective prediction consists of an AI-derived component, the contributions
209 of the players' initial predictions, and the stochastic AI error that remains after
210 aggregation.

211 Expected payoffs

212 We next derive the expected payoff under each incentive structure. Expectations are
213 taken over the distributions of the causal factors and stochastic AI errors, conditional
214 on the current population strategies and the fixed parameters of the prediction target.

215 Feedback

216 Under the feedback structure, a player's payoff depends on the alignment between
 217 their revised prediction and the residual error of the collective prediction. To express
 218 this residual, we first define the remaining error in the intercept and in the coefficient
 219 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S27})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S28})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S29})$$

220 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 221 remaining error associated with factor i . Because the causal factors and stochastic AI
 222 errors are independent and have zero mean, cross terms vanish when the expectation
 223 is taken. Substituting the revised individual prediction and collective residual into
 224 the feedback payoff gives

$$\mathbb{E}[\pi_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S30})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S31})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \quad (\text{S32})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S33})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S34})$$

225 The first term represents the alignment between the AI-derived prediction and the
 226 collective residual, whereas the second term represents the contribution of the player's
 227 initial prediction. The final term arises because the stochastic error in player A 's AI
 228 prediction also contributes to the collective prediction. This finite-population effect
 229 decreases as population size increases.

230 Niche-expert

231 Under the niche-expert structure, the payoff is the negative squared error of the
 232 revised individual prediction, weighted by the frequency of the player's interest.
 233 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_0(Y_A^{\text{re}} - Y)^2] \quad (\text{S35})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S36})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S37})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S38})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S39})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \quad (\text{S40})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S41})$$

$$= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B\right. \quad (\text{S42})$$

$$\left. + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S43})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S44})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2 \quad (\text{S45})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S46})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}[\pi_{A, \text{Feedback}}] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A, \quad (\text{S47})$$

$$\mathbb{E}[\pi_{A, \text{Niche-expert}}] = \mathbb{E}\left[-\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S48})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

247 Expected accuracy

248 Using the residual defined above, the mean squared error of the revised collective
249 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S49})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S50})$$

250 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y}^{re})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S51})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S52})$$

251 The collective prediction error consists of the squared error in the intercept, the re-
252 maining error associated with each causal factor, and the stochastic AI error remain-
253 ing after averaging across players. Counterfactual collective accuracy is obtained
254 from the same expression by setting $\beta_A = 0$ for all players while retaining their
255 evolved interests and beliefs.

256 Chatbot AI

257 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each
258 player's interest. This causes the information contained in AI predictions to depend
259 directly on the distribution of interests in the population. Here, we derive the collec-
260 tive prediction, expected payoffs, and expected collective accuracy under this form
261 of AI assistance.

262 Model

263 The prediction generated by Chatbot AI for player A depends on the causal factor
264 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S53})$$

265 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
266 The stochastic error follows a Gaussian distribution with mean zero and standard
267 deviation τ and is generated independently across players and prediction events. The

268 player's initial and revised predictions are given by Player A's initial and revised
 269 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S54})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S55})$$

270 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 271 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 272 tion remains specific to the single factor associated with the player's interest.

273 Aggregation

274 Because both the player's initial prediction and the Chatbot AI prediction contain
 275 information about the same causal factor, we use clustering aggregation. Revised
 276 predictions are first averaged among players interested in the same factor and then
 277 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S56})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S57})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon_A], \quad (\text{S58})$$

278 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 279 players in that set.

280 To simplify the collective prediction, we define the effective coefficient, weighted
 281 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S59})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S60})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S61})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon_A, \quad (\text{S62})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S63})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S64})$$

where $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$ denotes the set of represented interests.

Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S65})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S66})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S67})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}[\pi_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S68})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S69})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S70})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S71})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

302 Niche-expert

303 Under the niche-expert structure, the negative squared error of each revised indi-
 304 vidual prediction is weighted by the frequency of the player's interest. We consider
 305 $e_A = 0$ separately because these players make constant predictions rather than factor-
 306 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S72})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S73})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S74})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S75})$$

307 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 308 expected payoff is then

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S76})$$

$$= -\mathbb{E}[\rho_{e_A} [\left\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \right\} X_{e_A} \quad (\text{S77})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S78})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S79})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S80})$$

309 In both cases, the expected squared error consists of the mismatch between the
 310 player's belief and the true coefficient, contributions from causal factors omitted
 311 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 312 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 313 interest.

314 Incentivizing or penalizing AI use

315 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 316 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}[\pi_{A,Feedback}] = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S81})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S82})$$

$$\mathbb{E}[\pi_{A,Niche-expert}] = \mathbb{E} \left[-\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S83})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S84})$$

Balanced

Under the balanced structure, the payoff corresponds to the weighted mean of the feedback and niche-expert payoff components, with balancing weight w :

$$\mathbb{E}[\pi_A] = \mathbb{E} \left[(1 - w)Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - w\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] \quad (\text{S85})$$

$$= (1 - w)\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - w\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] \quad (\text{S86})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

Unrevised balanced

The unrevised feedback and niche-expert components are identical to those in the model without AI assistance (Wang et al., 2025). We therefore calculate their expected payoffs following the derivations in the original model and take their arithmetic mean:

$$\mathbb{E}[\pi_A] = \frac{1}{2}\mathbb{E} \left[Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2 \right]. \quad (\text{S87})$$

Combined balanced

Under the combined balanced structure, the payoff is the arithmetic mean of the balanced and unrevised balanced payoffs, with $w = 0.5$ for both balanced components. Its expected payoff is therefore

$$\mathbb{E}[\pi_A] = \frac{1}{4}\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right. \quad (\text{S88})$$

$$\left. + Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2 \right]. \quad (\text{S89})$$

The expected payoff under the combined balanced structure can therefore be calculated directly from the revised and unrevised feedback and niche-expert payoffs derived above.

335 Expected accuracy

336 Finally, we derive the expected collective accuracy under clustering aggregation.
 337 The intercept α_0 serves as the baseline prediction. Because the causal factors are
 338 independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S90})$$

339 Using the expression for the collective residual above, the expected squared error of
 340 the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S91})$$

341 The first two terms represent squared errors in the intercept and factor-specific coef-
 342 ficients. The final term represents the stochastic AI error remaining after predictions
 343 are averaged within each interest group. Thus, stochastic error has a larger effect
 344 when an interest is represented by fewer individuals. Normalizing this error relative
 345 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S92})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S93})$$

346 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
 347 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

348 Evolutionary dynamics

349 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,
 350 2005), we use replicator dynamics as a deterministic large-population approximation
 351 to characterize equilibrium states and their local stability. The agent-based sim-
 352 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff
 353 stationary states, the replicator and Fermi formulations imply the same local invasion
 354 directions for any finite selection strength $s > 0$, although their transient dynamics
 355 need not coincide. We focus on the feedback and niche-expert structures, for which
 356 the long-term dynamics can be characterized analytically. The direct interventions
 357 on AI use and the balanced structure are examined through simulations.

Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S94})$$

where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an internal stationary state, all strategies with positive frequency must have the same expected payoff. A boundary state can also be stationary because a strategy with zero frequency cannot increase unless it is introduced into the population. We therefore assess the stability of boundary states by asking whether they can be invaded by a rare alternative strategy.

Omniscient AI

With Omniscient AI, every AI prediction contains information about the full prediction target, and revised predictions are combined through averaging. As population size increases, independent AI errors vanish from the collective prediction. However, these errors remain in individual prediction errors and can therefore continue to affect niche-expert payoffs. We consider the two incentive structures separately.

Feedback

Under the feedback structure, both the stochastic error in the collective prediction, $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous work (Wang et al., 2025), we define the Lyapunov function as the expected error of collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S95})$$

The function V is non-negative and equals zero when the collective prediction is identical to the target. To determine how V changes over time, we take its time derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S96})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S97})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S98})$$

For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S99})$$

381 Although K changes with the population state, it is common to all strategies at a
 382 given state and can therefore be treated as a constant with respect to the strategy
 383 index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S100})$$

384 Under the replicator equation, the rate of change in the population mean of a
 385 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S101})$$

386 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S102})$$

387 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S103})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S104})$$

388 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
 389 λ is

$$u_{\lambda} = K\beta_{\lambda} + (1 - \beta_{\lambda})c_{\lambda}G_{e_{\lambda}}. \quad (\text{S105})$$

390 This expected payoff is identical to the quantity inside the covariance above. There-
 391 fore,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S106})$$

392 Therefore, the expected error of collective prediction cannot increase during the
 393 evolutionary process and decreases whenever strategies differ in their expected pay-
 394 offs. The derivative becomes zero when the variance of expected payoffs is zero,
 395 $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same
 396 expected payoff. However, this condition does not necessarily imply that $V = 0$,
 397 because a population can lack strategies that would improve its collective prediction.

398 If $V > 0$, at least one component of the collective residual remains nonzero. A rare
 399 strategy whose prediction is aligned with this residual can obtain a higher expected
 400 payoff than the resident strategies and invade the population. Thus, provided that
 401 novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$.
 402 At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

403 Niche-expert

404 Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each
 405 individual. Stochastic AI error therefore remains in the expected payoff even when
 406 $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

407 We first consider strategies having the same interest by optimizing the expected
 408 payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly
 409 concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S107})$$

410 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 411 unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S108})$$

412 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 413 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the
 414 boundary cases do not require a separate evolutionary argument; they only change
 415 the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the
 416 revised prediction is independent of c_e , so different values of c_e are selectively neutral
 417 and can be treated as a single payoff-equivalent strategy class.

418 To compare the optimized strategies across interests, let L_e denote the minimum
 419 expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal
 420 optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S109})$$

421 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 422 the expected squared prediction error at the corresponding boundary.

423 Interest 0 must be considered separately because $X_0 = 1$ and the player's belief
 424 determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S110})$$

425 Substituting this into the expected payoff gives a strictly concave function of β , with
 426 the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S111})$$

427 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S112})$$

428 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S113})$$

429 At this optimum, the player adjusts their belief to offset the systematic AI bias. The
 430 stochastic component of AI error cannot be removed in the same way and therefore
 431 limits the benefit of AI reliance.

432 We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different
 433 values of c_e are selectively neutral and can be treated as a single payoff-equivalent
 434 strategy class. A stable state can contain only the optimal strategy class within each
 435 represented interest because a suboptimal strategy can be invaded by S_e .

436 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 437 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S114})$$

438 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
 439 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \dots = \rho_M^* L_M. \quad (\text{S115})$$

440 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S116})$$

441 This internal stationary state is locally stable. At equilibrium, all represented in-
 442 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 443 interest increases slightly above its equilibrium value, its payoff decreases because
 444 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
 445 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
 446 namics therefore generate a restoring force that reduces small deviations in interest
 447 frequencies and drives the population back toward the internal stationary state.

448 If one or more interests are absent, the resulting boundary state can be stationary
 449 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 450 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 451 population is negative when $L_e > 0$, so the rare interest increases when introduced
 452 at sufficiently small frequency.

453 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 454 prediction strategies and their frequencies. Whether the constrained optimum has
 455 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 456 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 457 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 458 interest frequency remain uniquely determined.

459 Having characterized this equilibrium, we next compare its collective prediction
 460 with a prediction made entirely by AI. In particular, the expected squared error of
 461 an Omniscient AI prediction is

$$\mathbb{E}[(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S117})$$

462 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 463 fore, because L_e is the minimum individual expected squared prediction error within
 464 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S118})$$

465 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S119})$$

466 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S120})$$

467 The collective prediction is the average of individual revised predictions. By convex-
 468 ity of the squared error,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S121})$$

469 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{\text{AI}})^2]. \quad (\text{S122})$$

470 Thus, the diversity of optimized predictions allows the population to outperform an
 471 individual AI prediction at equilibrium.

472 However, outperforming AI does not imply perfect collective accuracy. Under
 473 averaging, the non-AI contribution associated with a particular factor is provided
 474 only by individuals interested in that factor. The predictions of individuals with
 475 other interests therefore attenuate that factor's coefficient in the collective prediction.
 476 For a nonzero factor i , the residual coefficient of the stationary collective prediction
 477 can be written as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S123})$$

478 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for
 479 every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S124})$$

480 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 481 Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S125})$$

482 and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-
 483 expert structure preserves interest diversity and allows the population to outperform
 484 AI, but averaging prevents the population from reaching perfect collective accuracy.

485 AI models under alternative aggregation rules

486 In the main analysis, we assume that collective predictions under the omniscient AI
 487 and chatbot AI models are generated using averaging and clustering aggregation,
 488 respectively. However, the aggregation rules can, in principle, be swapped between
 489 the two AI models. To examine the consequences of these alternative combinations,
 490 we investigate their behavior through mathematical derivation and simulation.

491 Omniscient AI with clustering aggregation

492 Under clustering aggregation, the collective prediction based on individual predic-
 493 tions revised by the omniscient AI can be written as

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S126})$$

$$= \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S127})$$

$$= \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\beta_A (Y + B + \epsilon_A) + (1 - \beta_A) c_A X_{e_A}], \quad (\text{S128})$$

494 where $\mathcal{A}_i$ is the set of players interested in factor i , $n_i = |\mathcal{A}_i|$ is the number of players
 495 in that set, and $\mathcal{I} = \{i \in \{0, \dots, M\} : n_i > 0\}$ is the set of represented interests.

496 To simplify the collective prediction, we define:

$$c_i^{agg} = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A, \quad (\text{S129})$$

$$\bar{\beta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A, \quad (\text{S130})$$

$$\beta^{agg} = \sum_{i \in \mathcal{I}} \bar{\beta}_i, \quad (\text{S131})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon_A, \quad (\text{S132})$$

$$\bar{\eta} = \sum_{i \in \mathcal{I}} \tilde{\eta}_i. \quad (\text{S133})$$

497 Thus, the collective prediction can be written as

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} c_i^{agg} X_i + \beta^{agg}(Y + B) + \bar{\eta}. \quad (\text{S134})$$

498 Because individual reliance on AI, β_A , is initialized independently from $U(0, 1)$,
 499 the expected initial value of β^{agg} is $|\mathcal{I}|/2$. When all $M + 1$ interests are represented,
 500 this becomes $(M + 1)/2$. Thus, clustering aggregation incorporates the omniscient AI
 501 prediction repeatedly across interest groups, scaling its contribution to the collective
 502 prediction by a factor of approximately $(M + 1)/2$ at initialization. For a large number
 503 of factors, such as $M = 50$, this produces an unrealistically large contribution of the
 504 AI prediction to the collective prediction. Therefore, clustering aggregation does not
 505 provide a reasonable aggregation rule for the omniscient AI model in the context
 506 of our study, as it repeatedly incorporates predictions of the entire outcome across
 507 interest groups.

508 Chatbot AI with averaging aggregation

509 Unlike omniscient AI with clustering aggregation, chatbot AI can be combined with
 510 averaging aggregation without an apparent mismatch in prediction scale. Under av-
 511 eraging aggregation, the collective prediction based on individual predictions revised
 512 by the chatbot AI can be written as

$$\hat{Y}^{re} = \frac{1}{N} \sum_{A=1}^N Y_A^{re} \quad (\text{S135})$$

$$= \frac{1}{N} \sum_{A=1}^N [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon_A]. \quad (\text{S136})$$

513 We define the aggregate revised coefficient associated with each factor, the ag-
 514 gregate AI bias, and the stochastic error as

$$c_i^{av} = \frac{1}{N} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_i + (1 - \beta_A) c_A\}, \quad (\text{S137})$$

$$B^{av} = \frac{1}{N} \sum_{A=1}^N \beta_A b_{e_A}, \quad (\text{S138})$$

$$\bar{\eta} = \frac{1}{N} \sum_{A=1}^N \beta_A \epsilon_A. \quad (\text{S139})$$

515 Thus, the collective prediction becomes

$$\hat{Y}^{re} = \sum_{i=0}^M c_i^{av} X_i + B^{av} + \bar{\eta}. \quad (\text{S140})$$

516 We define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - c_0^{av} - B^{av}, \quad (\text{S141})$$

$$\delta_i = \alpha_i - c_i^{av}, \quad (i \neq 0). \quad (\text{S142})$$

517 Expected payoffs

518 Under the feedback structure, the expected payoff is

$$\mathbb{E}[\pi_A] = \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{\beta_A^2 \tau^2}{N}, \quad (\text{S143})$$

519 where $\sigma_0^2 = 1$.

520 Because the niche-expert payoff depends only on individual prediction error and
521 the frequency of the player's interest, its expected payoff is unaffected by the ag-
522 gregation rule and is identical to that derived for the chatbot AI with clustering
523 aggregation.

524 Expected accuracy

525 The expected squared error of the collective prediction is

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{A=1}^N \beta_A^2. \quad (\text{S144})$$

526 Therefore, the expected collective accuracy is

$$R = 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{A=1}^N \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}. \quad (\text{S145})$$

527 Simulation results indicate that, under averaging aggregation, chatbot AI has a
528 limited effect on the collective dynamics (Fig. S10). Under the feedback structure,
529 collective accuracy reaches one across AI bias levels, while median reliance on AI
530 shows little variation with AI bias. In contrast, with the niche-expert incentive,
531 collective accuracy remains near zero, as in the absence of AI assistance, despite
532 relatively high reliance on AI at some levels of AI bias.

Supplementary Figure

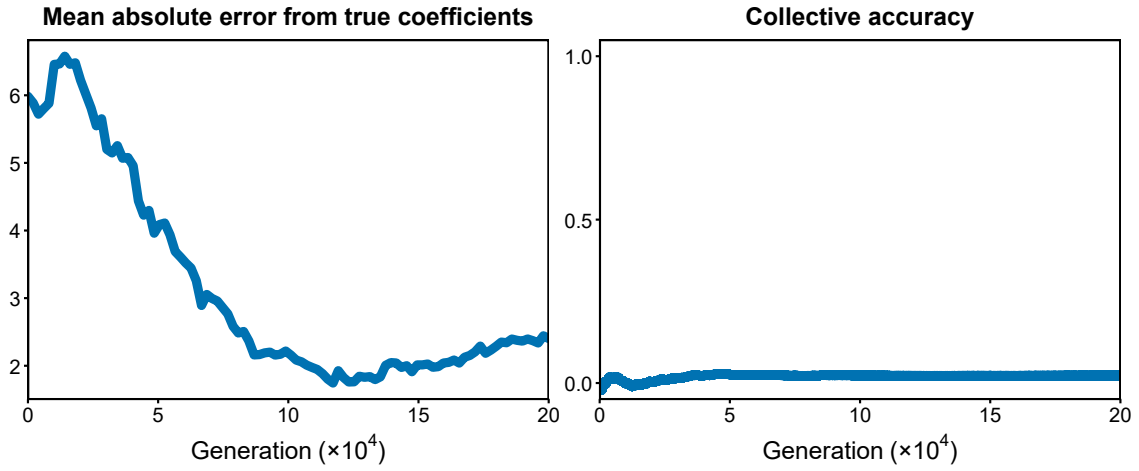


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $M + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

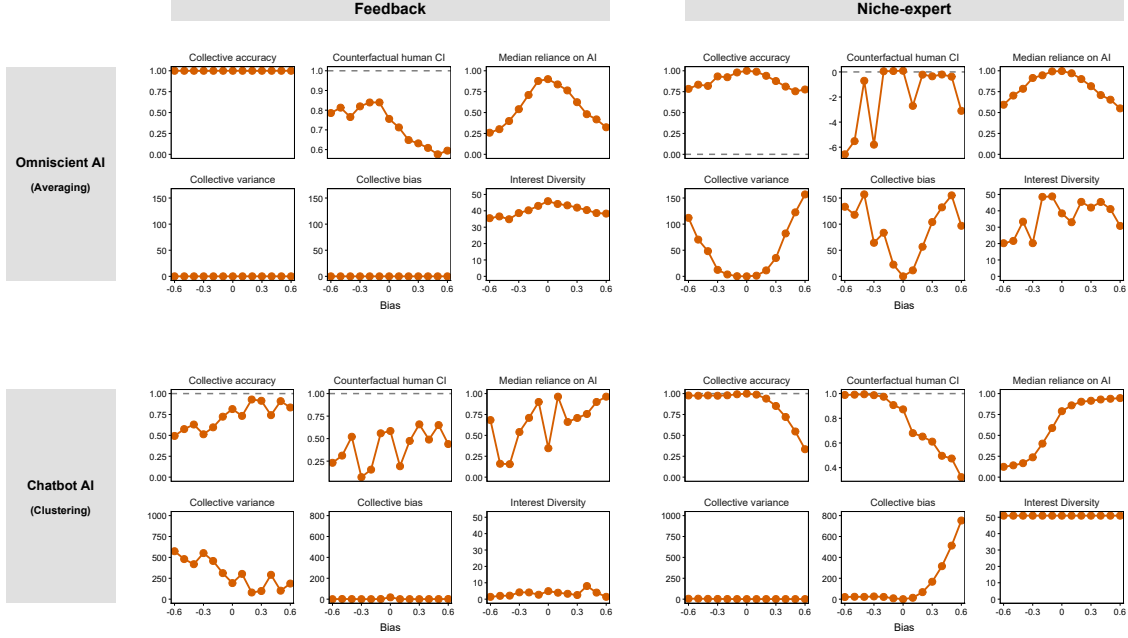


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

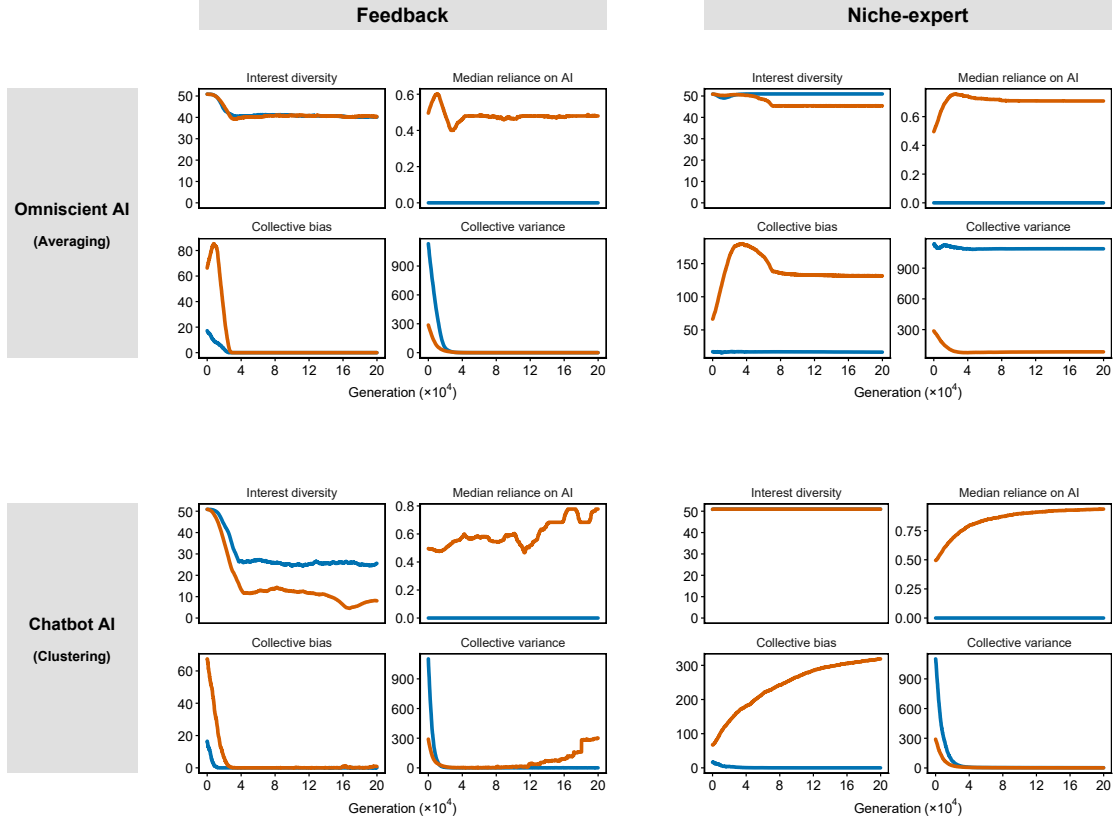


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

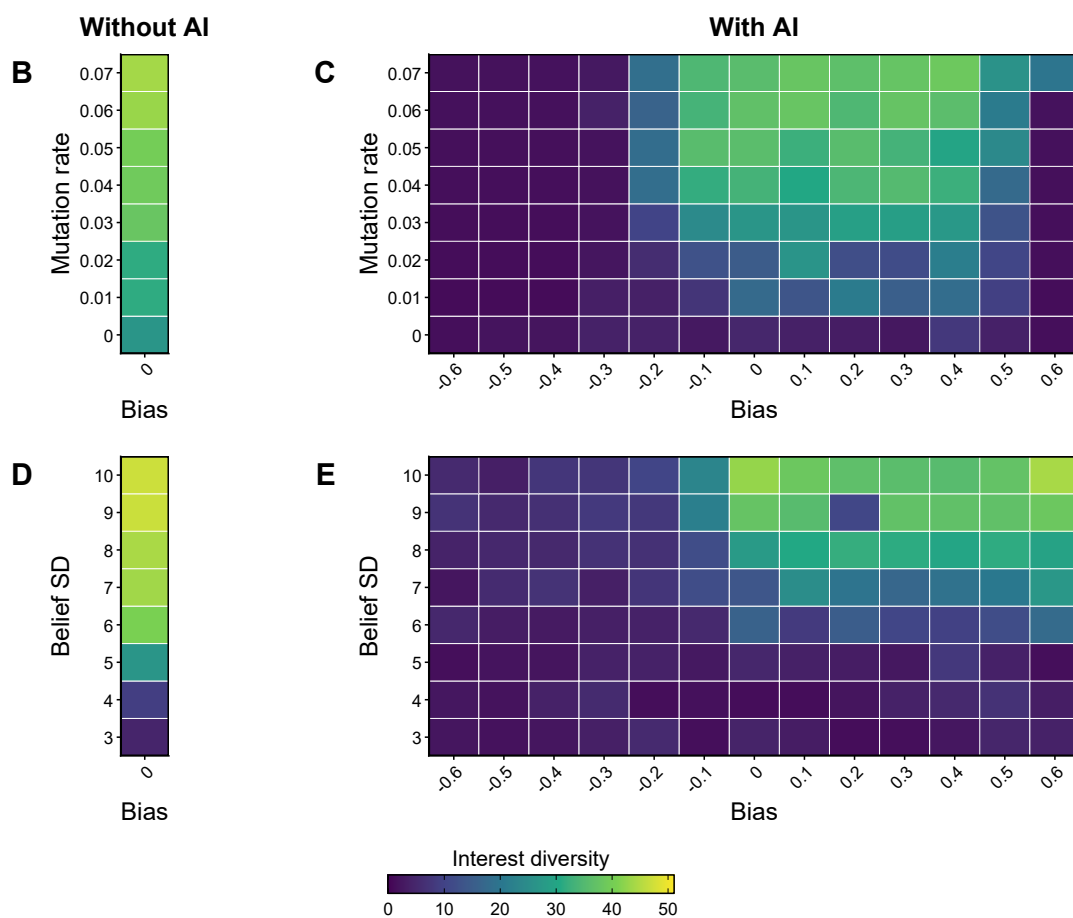
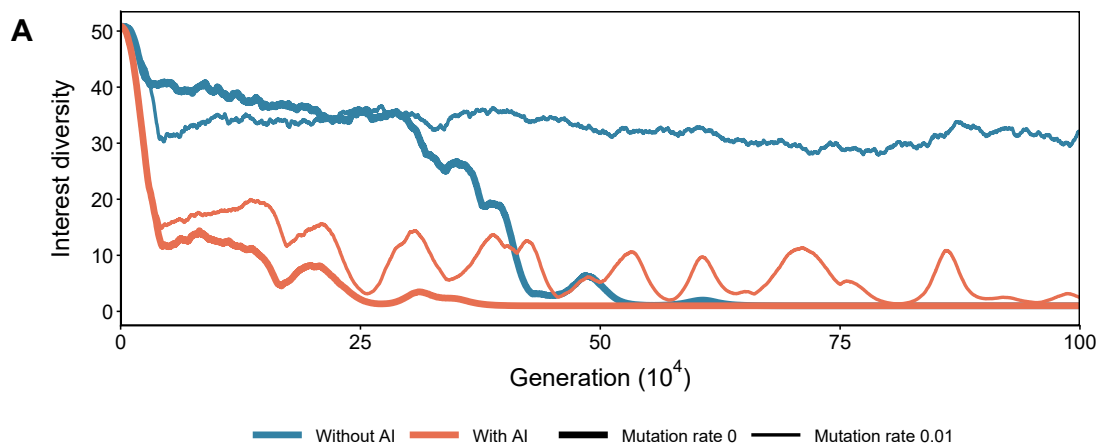


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

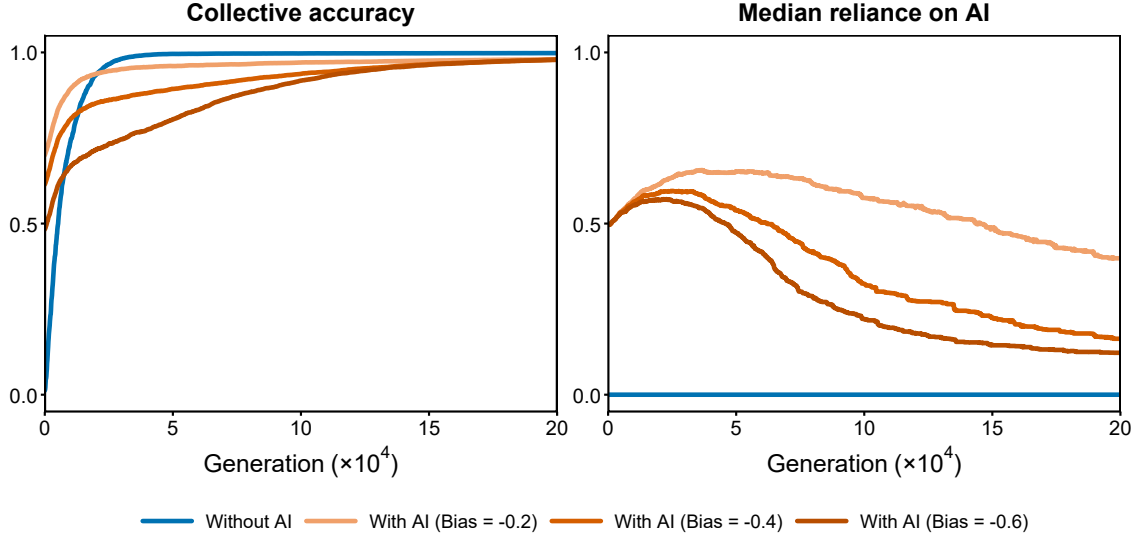


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

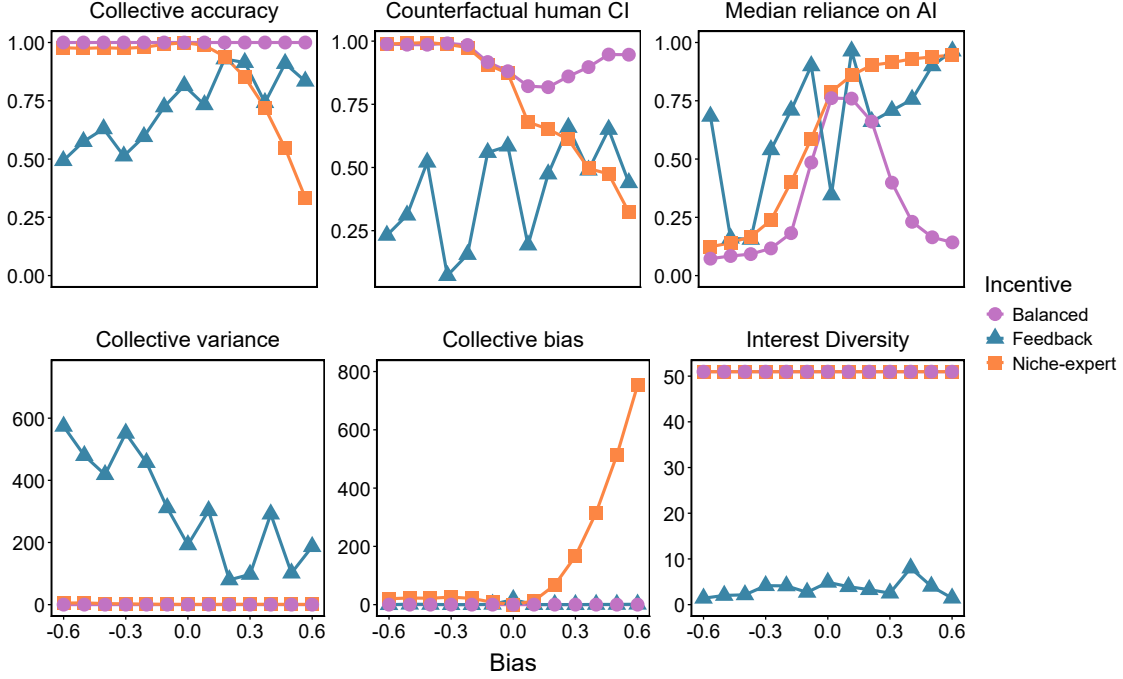


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,001–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

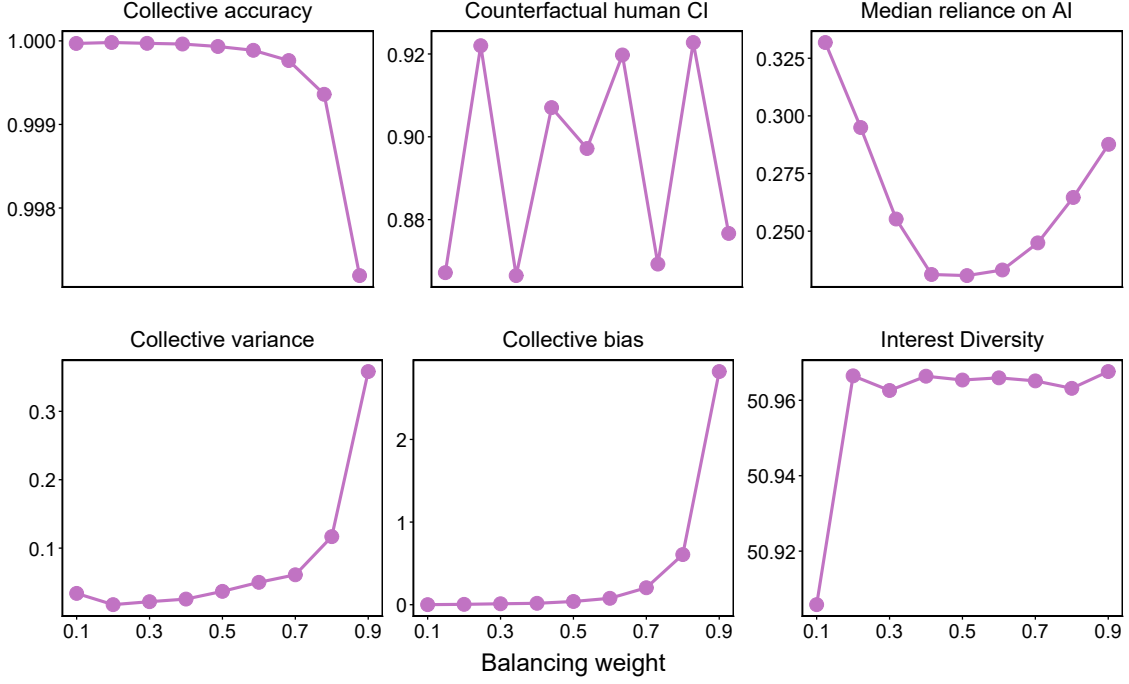


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

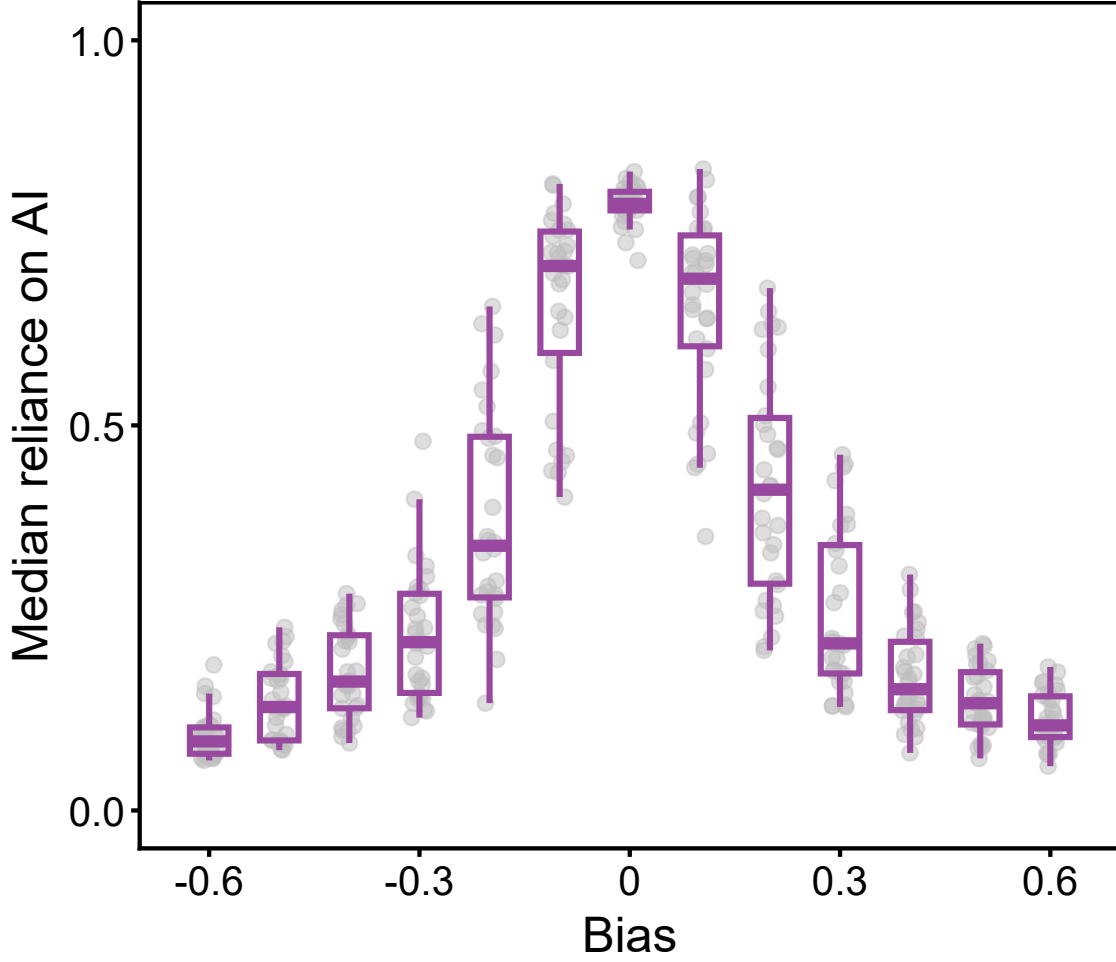


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

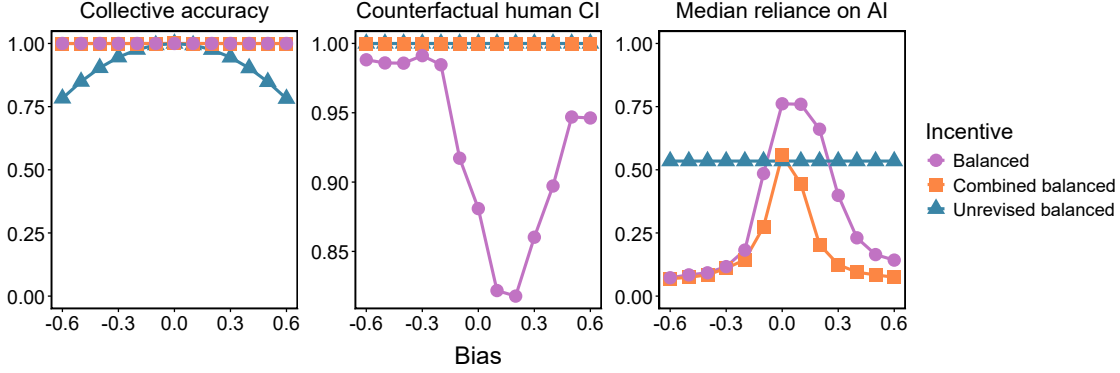


Figure S9: **Incentive structures incorporating unrevised prediction accuracy can promote better counterfactual human CI.** Evaluation metrics under the balanced, unrevised balanced, and combined balanced structures across varying levels of AI bias. The unrevised balanced structure is defined as the arithmetic mean of the feedback and niche-expert structures based on unrevised predictions, whereas the combined balanced structure is defined as the arithmetic mean of the balanced and unrevised balanced structures. All evaluation metrics are averaged over generations 190,001–200,000. The unrevised balanced structure achieves near-perfect counterfactual human CI; however, median reliance on AI remains near 0.5 because the unrevised balanced incentive does not account for AI predictions and therefore imposes no selection on β_A . Consequently, collective accuracy directly inherits AI bias and decreases as AI bias increases. The combined balanced structure, which leverages both the balanced and unrevised balanced structures, achieves near-perfect counterfactual human CI without compromising collective accuracy. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$; Chatbot AI is used.

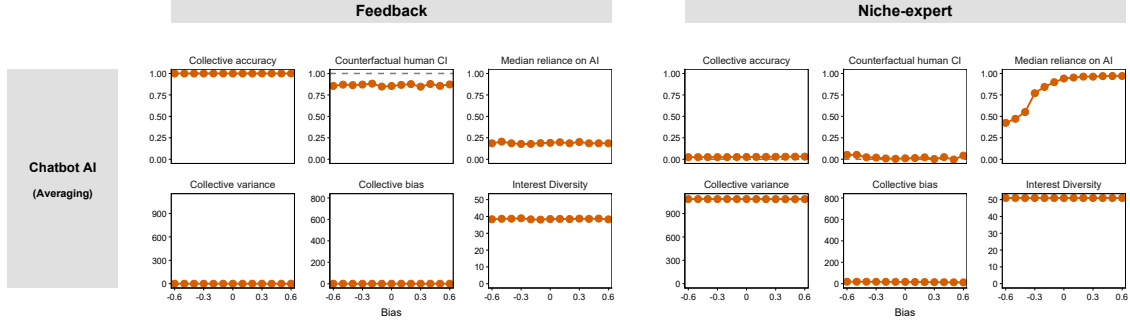


Figure S10: **Chatbot AI with averaging aggregation has a limited influence on CI.** Evaluation metrics across varying levels of AI bias under Chatbot AI with averaging aggregation. Two incentive structures, feedback and niche-expert, are considered. A larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, whereas counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as the Hill–Shannon diversity of interests within the population, and median reliance on AI represents the median reliance on AI (β) across individuals. Collective bias represents the systematic deviation of the collective prediction from the true target, whereas collective variance represents the variability of collective prediction errors across prediction events. All evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

Table S1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

534 References

- 535 Hong, L., S. E. Page, and M. Riolo (2012). Incentives, information, and emergent
536 collective accuracy. *Managerial and Decision Economics* 33(5-6), 323–334.
- 537 Mann, R. P. and D. Helbing (2017). Optimal incentives for collective intelligence.
538 *Proceedings of the National Academy of Sciences* 114(20), 5077–5082.
- 539 Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game
540 dynamics. *Mathematical biosciences* 40(1-2), 145–156.
- 541 Traulsen, A., J. C. Claussen, and C. Hauert (2005). Coevolutionary dynamics: from
542 finite to infinite populations. *Physical review letters* 95(23), 238701.
- 543 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Individual incentives that
544 promote collective intelligence. *Proceedings of the National Academy of Sci-*
545 *ences* 122(51), e2516535122.